Constants as the bridge

Minimal-shot autonomy, in motion

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A snow plough's job is short: keep the road clear. The catch is that, while the plough is doing it, the road is invisible. Curbs are buried, lane markings are gone, the seam between asphalt and garden is no longer drawn. A self-driving stack trained on Cityscapes will report, with calibrated confidence, that the entire scene is sky.

The familiar response is *we just need more data*. Annotate snowy roads. Annotate dust storms. Annotate fog, lava, washouts. There are 27 million miles of road in the world, and the long tail of conditions any of them can be in is longer than the road itself. We are not going to label our way out of it.

There is a different move available. For almost every operating regime where autonomy fails for lack of data, there is an adjacent regime — temporally, seasonally, geographically — where we have plenty of data, and where the parts that matter are the same. A snow plough's road is the same road it was last July. The curb hasn't moved. The hydrant hasn't moved. The road's *appearance* has changed completely; its *position in space* has not.

If we can identify what stays constant between the data-rich regime and the data-poor one, we can extend our existing models into the new regime without learning a single thing about it. We use the constants as a bridge.

This essay is one concrete demonstration of that idea, applied to autonomous snow ploughs. The principle is general; the snow plough is the vehicle. The headline artefact is a video: a continuous overlay of road position on a snow-buried street, frame by frame, produced by a pipeline whose only learned components have never seen snow.

The example

Self-driving systems are trained on dry roads, deliberately. Cityscapes, KITTI, nuScenes, Waymo Open — the canonical training corpora — are dominated by clear-weather European or American highways under daylight. A snow plough operates in the regime that data deliberately excludes. Mistakes there damage infrastructure. The familiar move — collect more snowy data, annotate it, retrain — is uneconomic at the scale of a 27-million-mile road network.

But the road's *location* hasn't moved. For almost every road in the developed world there is a clear-season image of it: Street View, Mapillary, the operator's own prior captures, the dataset's own summer traversal. The plough's missing information is not what a road looks like under snow. It is *where this road sits in the camera frame right now*. That is a registration problem, not a learning problem. We solved registration twenty years ago.

So the system, in six steps:

1. Pull the live snowy frame from the plough's camera.
2. Pull a clear-season prior of the same coordinates. We use the Boreas dataset's matched summer drives for the canonical demo, and Mapillary for the static-stills precursor.
3. Match the two images using a generic, frozen feature matcher. The matcher anchors on what stays constant between the two: buildings, signs, poles, rooflines, distant horizons.
4. Estimate a homography from the matches, biased toward the ground plane.
5. Run a generic Cityscapes-trained road segmenter on the clear prior — never on the snow frame.
6. Warp the road mask through the homography onto the snowy frame.

The plough now knows where the road is, and where it isn't, even though it cannot see the road, and even though no model in the pipeline has ever been trained on a snowy frame.

Architecture

DISK (Tyszkiewicz et al., NeurIPS 2020) extracts local features. LightGlue (Lindenberger et al., ICCV 2023) matches them. USAC-MAGSAC (Barath et al., CVPR 2020) fits a homography by RANSAC, restricted to lower-image matches to bias toward the ground plane. Mask2Former (Cheng et al., CVPR 2022), pretrained on Cityscapes (Cordts et al., CVPR 2016), produces the road mask on the clear prior. The mask (and its warp into snow image space) is reduced to its single largest connected component, because a plough cares about the *one* drivable surface in front of it.

The video extension wraps that static pipeline in three thin layers:

- A track loader indexing a snow stream and a paired summer stream by GPS pose.
- A prior pool that, for each snow frame, picks the K = 3 nearest summer captures by UTM distance and caches their Mask2Former mask once.
- An EMA temporal smoother ($\alpha = 0.4$) running over the binary mask, suppressing per-frame jitter without introducing the drift we saw with optical-flow propagation.

A pickled cache layer makes the matching pass reusable: subsequent renders that change only the smoother (`temporal=ema|flow|none`) skip the ~50-minute matching pass entirely.

Snow imagery comes from Boreas (Burnett et al., IJRR 2023), CC BY 4.0 on the AWS Open Data registry — same FLIR Blackfly S camera, same Glen Shields loop, with paired summer traversals on the same UTM coordinates. Mapillary remains the open-imagery substrate for the static-stills precursor.

Every learned component is frozen. Snow appears only at inference, as the runtime input.

What we showed

A 15-second clip from a snow-buried Toronto residential street (Boreas `boreas-2021-01-26-11-22`, January 2021) with a continuous green road overlay tracking the buried road frame by frame. Side by side with the naive baseline — the same Cityscapes segmenter applied directly to the snow frame, painting a confident red mask over the entire scene — the contrast is the demonstration. The naive method's red drifts. The cross-season pipeline's green stays on the road.

We render the same clip in five layouts (single overlay, snow-vs-overlay sidebyside, two 3-panel orderings with the naive baseline, and a 2 × 2 quad with the summer prior visible) so the audience can see the same evidence at different depths. The single overlay is the headline; the quad is the receipts.

The static-stills precursor is preserved in the same repository (`make stills`) and produces fourteen user-curated GREAT/OKAY hero pairs over Northern Sweden and Finland — the v1.x demo's foundation, on which the video extension was built.

The most interesting honest finding survives from the static work and into the moving demo: inlier count is not a reliable predictor of overlay quality. A pair with hundreds of inliers can warp the mask onto the wrong region if the inliers concentrate on building façades. A pair with seventeen inliers can be perfect if those seventeen happen to land on the road. The system therefore needs a human in the loop on the input and the output, even after the matcher succeeds.

What we extended (and what we tried that didn't work)

The video pipeline is a thin wrapper around the static pipeline, but two extensions deserve note for what they reveal.

We tried using previous snow frames as additional priors for the current frame ("synthetic priors"). Snow→snow matching is much higher confidence than snow→summer because lighting, lens, and viewpoint conditions are identical between consecutive snow frames; the matcher returned three to seven times more inliers per pair. In single-frame stills, the resulting mask was visibly broader and more confident. In motion, the mask drifted outward over time: each frame's slightly-too-large mask seeded the next frame's synthetic prior, which produced a slightly-larger mask, and the road mask grew into bushes and treelines over a few seconds. The failure was a positive feedback loop the static stills hid. We rejected synthetic priors.

We tried optical-flow propagation between matched keyframes. Same outcome, different mechanism: forward-driving cameras have vanishing-point flow that stretches the previous mask outward at every step. We rejected it.

We kept EMA on the binary mask, $\alpha = 0.4$. It is the simplest possible smoother and it does the least damage: it drops jitter without drifting, and on a frame where matching fails entirely, it holds the previous good mask rather than flickering empty.

This pattern — counterintuitive failures of "obvious" improvements that look better in stills — is itself an artefact of the demo. Static frames hide motion artefacts. We learned to verify motion before claiming wins.

What we didn't

We didn't train anything. We didn't fine-tune. We didn't collect a snow corpus. We didn't write a single line of snow-aware logic. The only handle we offered the model on the snow regime was the clear prior of the same place, plus the generic robustness of pretrained matchers and segmenters that have never seen snow.

We didn't claim the system replaces lidar or 3D depth estimation. The output is a 2D road *prior*, not a drivable-surface estimate. We didn't claim the homography is exact — it isn't, the world isn't planar — only that it transfers the road mask approximately and the approximation is enough.

We didn't claim real-time. The current matcher runs at ~5 s per snow→summer pair on Mac CPU; the canonical 15-second clip's matching cache builds in ~50 minutes. The cache layer makes that cost amortise across renders, but real-time would require a substantially faster correspondence model — that would be a learned component, and trained on synthetic snow to preserve the no-snow-in-training guarantee.

We didn't claim novelty in any single component. The novelty, such as it is, is in the *composition*: the matcher, the segmenter, and the homography are off-the-shelf; the move is using them together to bridge a regime where one of them would otherwise fail. The video extension adds GPS-keyed prior selection, EMA temporal smoothing, and a cache. None individually novel. The composition is.

Generalising

The structure of the move is: a model trained on regime A; an inference target in regime B; a known correspondence between the two; transfer through the correspondence. Snow on a road is one instance. Others admit the same structure: low-light medical imaging without low-light training data, polar earth observation without polar training data, a manipulator on Mars without Mars training data. In each case there is a regime in which we have plenty of data, an adjacent regime in which we don't, and a constant between the two — temporal, geometric, anatomical — that lets us bridge.

We are not going to label our way out of every long-tail regime. But for many of them, we don't have to. We just have to find what stays the same and walk across.

Code, video clips, walkthrough notebook, and the static-stills precursor: see the project repository. Reproducible from a clean clone via `make reproduce` *(canonical 15-second clip) or* `make stills` *(static-prior 14-pair Mapillary precursor). Boreas dataset (UTIAS-ASRL) under CC BY 4.0. Submission video composed externally; storyboard at* `docs/slides.md`*. Submitted to SoTA Commission I — Minimal-Shot Autonomy, May 2026.*